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Viability and Impacts of Low-carbon Fuels for Maritime Applications

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This study evaluates the lifecycle viability of ammonia, hydrogen, methanol, and Fischer–Tropsch diesel as low-carbon fuels for maritime shipping by integrating regional supply chain modeling with the NavigaTE global fleet model. Using this framework, we consider scenarios involving widespread adoption of single alternative fuels and production pathways to illustrate the big-picture advantages and disadvantages of each in the context of the global maritime industry. Production pathways that use renewable electrolytic or biogenic hydrogen offer the most favorable trade-offs between costs and emissions in most regions, achieving up to 90% and 80% reductions in well-to-wake emissions relative to low-sulfur fuel oil, respectively. However, each comes with substantial feedstock constraints. Electrolytic fuels require up to 6,800 TWh/year of electricity, equivalent to 23% of current global generation. Biogenic routes demand up to 1.6 Gt of lignocellulosic biomass, exceeding estimated sustainable availability from forestry residues. Blue hydrogen pathways that pair methane reforming with carbon capture and storage can reduce emissions below LSFO levels if sequestration is permanent, but require up to 24% of global natural gas supply and come at higher cost than electrolytic and biogenic options in many regions. Methanol and FT diesel production also necessitate large CO_2 inputs – up to 580 Mt and 1,300 Mt, respectively, which would require a major scale-up of carbon capture and utilization. Emissions and costs vary by up to a factor of ten depending on the production region, with the most viable options concentrated in countries offering low-cost, low-carbon electricity or natural gas.

KEY WORDS: Maritime Industry and the Environment; Maritime Decarbonization; Alternative Fuels; Low-carbon Fuels; Lifecycle Emissions

INTRODUCTION

Maritime shipping is the backbone of global trade, carrying over 80% of international cargo while contributing roughly 3% of global greenhouse gas (GHG) emissions (IMO 2021). Today, the sector relies almost entirely on fossil fuels to meet its energy needs (IEA 2023), making decarbonization both an urgent priority and a formidable challenge. In recognition of this, the International Maritime Organization (IMO) has approved a regulatory framework designed to achieve net zero emissions in the shipping industry by or around 2050, helping to catalyze research and investment into alternative fuels and propulsion systems (IMO, 2025).

A wide range of candidate fuels has emerged with potential to reduce maritime shipping emissions. In a comparative assessment of greenhouse gas (GHG) reduction strategies, Bouman et al. (2017) found that low-carbon biofuels offered the greatest potential for emissions reductions among a suite of technical and operational measures, including efficiency improvements, wind-assisted propulsion, and speed optimization. While that analysis focused only on biofuels and liquified natural gas (LNG), subsequent studies have expanded the scope to include a broader array of alternative fuels and production pathways. Ammonia, hydrogen, methanol, LNG, and drop-in biofuels are among the most extensively studied options. Methanol and ammonia benefit from relatively mature production infrastructure, but ammonia suffers challenges to onboard storage due to its toxicity, corrosivity, and low energy density (Bilgili 2023). Hydrogen introduces a range of new technical and logistical challenges related to storage, distribution, and vessel integration (Foretich 2021, ICCT 2021, Stolz 2022). Dimethyl ether (DME), though less frequently examined, offers advantages such as low particulate emissions and compatibility with diesel engines, but suffers from low energy density and limited current production capacity (Xing 2021, Bilgili 2023). Biofuels, produced from first-generation crop feedstocks or more advanced lignocellulosic and algal sources, continue to receive attention for their potential lifecycle CO_2 reductions, relatively low cost, and suitability as transitional low-carbon drop-in fuels (Foretich 2021). However, concerns remain regarding land use change, feedstock availability (Kalt 2020, Daehn 2025), and the relatively high costs of second- and third-generation biofuels (Bilgili 2023).

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Each of these fuel options carries distinct trade-offs across emissions, costs, scalability, and technical compatibility. LNG, for instance, provides modest CO_2 benefits but suffers from upstream methane slip, diminishing its climate advantages relative to other fuels (Bouman 2017, Foretich 2021). Meanwhile, the use of onboard carbon capture and storage could serve as a near-term solution to decarbonize fossil fuel powered shipping, but challenges remain around techno-economic factors including installation and operational costs and CO_2 storage (Zanobetti 2024, Charalambous 2025). Battery-powered shipping holds promise for coastal and short-sea routes – particularly fixed-route ferry services – but existing studies find this solution unsuitable for deep-sea voyages due to cargo loss and range limitations (Kistner 2024, Kersey 2022, Stolz 2022, Korberg 2021).

Law et al. (2021) highlight the importance of considering not only the fuel, but the pathway by which the fuel is produced and used onboard. The study compared costs and emissions of conventional heavy fuel oil and LNG with hydrogen, methanol, ammonia, and diesel across 12 production pathways, including the use of blue and electrolytic hydrogen, and biomass-derived methanol and diesel. Their results showed that differences in well-to-wake (WTW) costs and emissions between production pathways are often more important than differences between fuels. For instance, WTW costs for hydrogen, ammonia, and methanol using the same hydrogen pathway were found to vary by up to 100%, while producing the same fuel via blue vs. electrolytic hydrogen could result in a 200% cost difference.

Building on vessel-level assessments, global techno-economic models including GloTraM (Smith 2019) and NavigaTE (Hintz 2023) have been developed to model the operation and evolution of the global fleet with the uptake of alternative fuels and technologies. These global models enable insights into the overall impacts, costs, and feasibility of alternative fuel and technology adoption scenarios, and to project the future evolution of the global fleet composition and fuel sources subject to market forces, regulatory measures, and feedstock constraints (MMMCZCS 2021, Smith 2016).

Underlying the global trends, key fuel production parameters such as feedstock availability, electricity prices, and grid carbon intensity can also differ dramatically across countries, shaping the viability of hydrogen-based fuel pathways (IRENA 2019). A 2020 report by the Hydrogen Council projected that electrolytic hydrogen could be produced for as little as USD 1.90/kg by 2025 in regions with abundant low-cost renewables such as Chile, Australia, and Saudi Arabia, compared to USD 6/kg in northern Europe (Hydrogen Council 2020). Fasihi and Breyer (Fasihi 2020) similarly found that hydrogen production costs from solar photovoltaic (PV) and wind powered systems can vary by a factor of two depending on location. Demirhan et al. (2018) demonstrated that ammonia production costs via biomass, wind, and solar are highly sensitive to regional feedstock prices and availability, highlighting these dynamics through case studies in selected U.S. states.

Across techno-economic and operational criteria, the literature highlights several broad conclusions regarding fuel selection and deployment strategy. Methanol is consistently viewed as a strong near-term option for deep-sea shipping, combining relatively low infrastructure barriers with scalable production routes from renewable electricity and CO_2 (Xing 2021). Hydrogen and ammonia are frequently emphasized for their long-term decarbonization potential, particularly in domestic and short-sea contexts, though their adoption depends on advances in fuel handling, safety, and infrastructure (Foretich 2021). Biofuels are seen as transitional solutions thanks to their compatibility with existing systems, though long-term scalability is limited by feedstock availability.

This study evaluates the viability and impacts of four alternative maritime fuels – LNG, ammonia, methanol, and Fischer-Tropsch diesel – by modeling the WTW costs, emissions, and feedstock requirements to operate the global shipping fleet on each fuel. Rather than focusing on single-vessel comparisons, this analysis models the operation of the full global fleet for each fuel and broadens the range of production pathways considered relative to prior studies. These pathways include hydrogen production via low-temperature electrolysis, methane reforming with and without carbon capture and storage (CCS), and biomass gasification. Feedstock sources span water, fossil natural gas (NG), electricity from wind, solar, and country-level grids, and carbon from direct air capture (DAC), bioenergy, and CCS-equipped fossil pathways. This comprehensive treatment enables detailed comparisons between long-term decarbonization pathways and shorter-term transition options, highlighting how feedstock and production choices influence costs, emissions performance, and scalability.

By applying a consistent techno-economic modeling framework across 38 countries, the study captures how regional variations in energy and feedstock costs affect the relative cost and emissions performance of each pathway. Scaling feedstock demands to the global fleet also enables identification of potential resource constraints – such as global electricity, biomass, CO_2, and natural gas requirements – that could limit the scalability of different production options. In doing so, the approach anticipates infrastructure and resource bottlenecks that must be addressed to support a global transition. Together, these results provide a robust foundation for informing regionally differentiated fleet planning and fuel adoption strategies in the maritime sector.

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METHODOLOGY

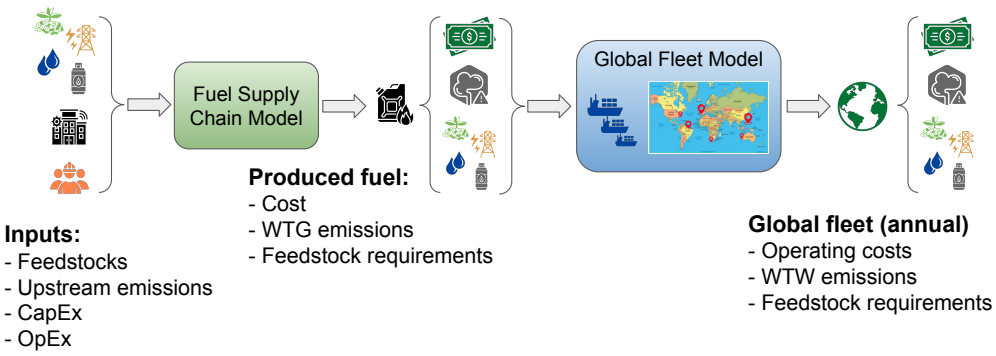


Fig. 1: Overview of inputs and outputs of the fuel supply chain and global fleet models.

Fig. 1 illustrates each component of fuel production and end use modeled in the study, and how they interface together. Outputs from the fuel supply chain model are fed into the global fleet model, which leverages the NavigaTE model developed by the Maersk Mc-Kinney Møller Center for Zero Carbon Shipping. The global fleet model calculates the total costs, emissions, and feedstock demands to operate the global fleet on the given fuel.

Fuel Supply Chain Model

The fuel supply chain model estimates well-to-gate (WTG) production costs, emissions, and feedstock demands for ammonia, methanol, Fischer-Tropsch (FT) diesel, LNG, and LSFO across 38 countries – selected for their global coverage and data availability – for each set of feedstocks and production pathways considered. The model evaluates present-day estimates for energy and feedstock inputs, process-specific emissions, capital and operating costs, and upstream supply chain impacts for each pathway, with results expressed per unit of fuel mass. Fig. 2 summarizes the feedstock processing and fuel production pathways modelled. Outputs from this model serve as inputs to the NavigaTE global fleet model (Hintze 2023) to assess fleet-wide impacts. Fuel production pathways are categorized as described in Table A.1.

The model integrates bottom-up production data from Aspen Plus process models and technoeconomic modeling based on the National Renewable Energy Laboratory's Hydrogen Analysis (H2A) framework with regionally-resolved energy prices, labor rates, and emissions intensities. Each feedstock processing or fuel production process p is characterized by the demand $D_{f,p}$ and country-level price $P_{f,c,p}$ and upstream emissions intensities $\varepsilon_{f,c,p}$ per unit of each input feedstock f, along with various processing costs and emissions. Figs. 3 and 4 compare WTG costs and emissions for ammonia across the modelled production pathways.

Feedstocks include electricity (grid, solar, or wind), NG, LCB, and water. Country-specific prices $P_{f,c,p}$ and emission factors $\varepsilon_{f,c,p}$ are obtained from literature. Electricity prices and emission rates are obtained for each country's grid from GlobalPetrolPrices.com and Our World in Data (Ember and Energy Institute 2024), respectively, with data for missing countries obtained from the UK Department for Energy Security and Net Zero (UK DESNZ 2024), and Electricity Maps (Electricity Maps 2022). The levelized cost of electricity (LCOE) for solar PV and onshore wind is primarily sourced from IRENA (IRENA 2024), which provides recent country-level data for most regions of interest. Where IRENA data is unavailable, values are supplemented using EMBER's 2024 ASEAN report (Dinita 2024) and the IEA's Levelised Cost of Electricity Calculator (IEA 2020). For countries without data, an extrapolation procedure is used to estimate their LCOE based on countries that are nearby or have similar characteristics. Global estimates of embedded CO_2e emissions from the construction of solar and wind power are obtained from Gan et al. (2024). For NG, a global estimate of upstream emissions, water, and electricity demand for NG recovery and processing, along with additional natural gas used for these processes, is evaluated using GREET 2024 (ANL 2024). NG prices are obtained from GlobalPetrolPrices.com, with data for missing countries obtained from the South Africa Department of Mineral Resources and Energy (DMRE 2023), Eurostat (Eurostat 2025), and Intratec (Intratec 2025). A global price and upstream emission rate for LCB is estimated from Dutta et al. (2011) and GREET 2024, respectively. Country-level water prices are obtained from the International Benchmarking Network for Water & Sanitation Utilities (IBNET 2025).

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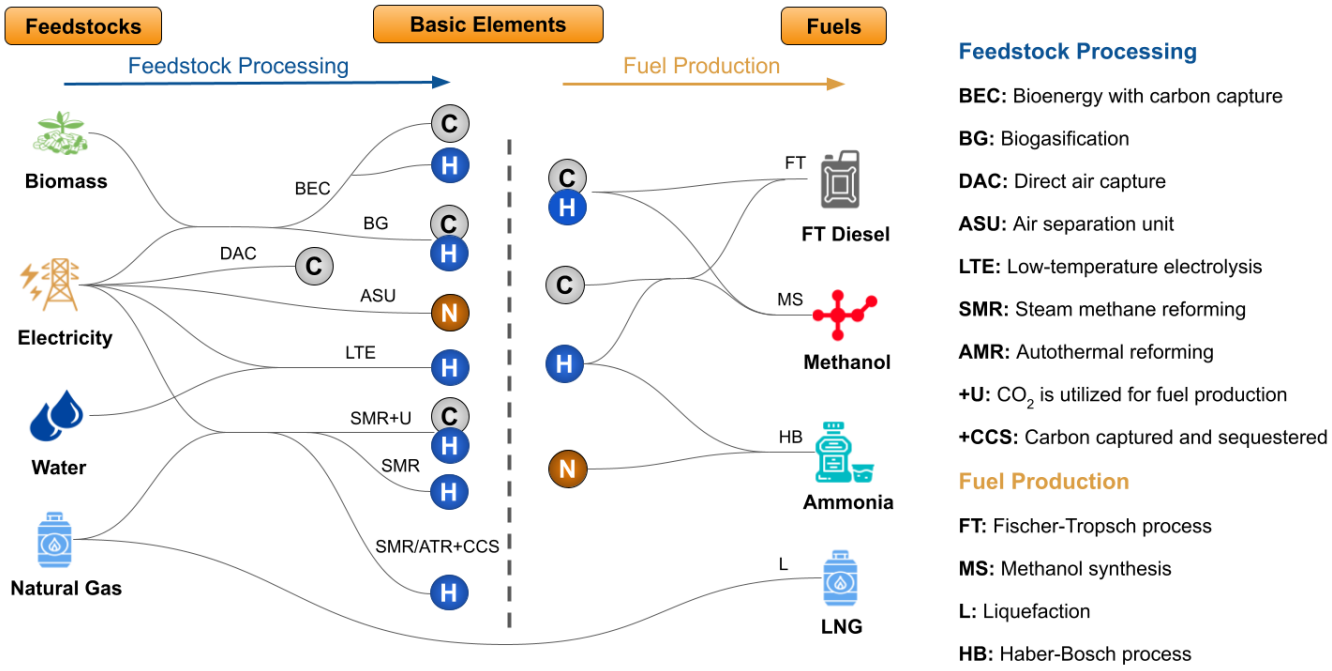


Fig. 2: Illustration of the fuel production pathways considered in the fuel supply chain model.

Cost modeling applies country-specific installation factors to scale U.S.-based capital costs to other countries:

$$\mathrm{CapEx}_{c,p} = \mathrm{BaseCapEx}_{\mathrm{USA},p} \times \mathrm{InstallFactor}_{c} \quad (1)$$

Operating expenditures (OpEx) are separated into fixed (fOpEx) and variable (vOpEx) components:

$$\mathrm{fOpEx}_{c,p} = \frac{W \cdot L_{c,p} \cdot N_{\mathrm{emp},p}}{Y_p} \cdot (1 + r_{\mathrm{GA}}) + (r_{\mathrm{OM}} + r_{\mathrm{Tax}}) \cdot \mathrm{CapEx}_{c,p} \quad (2)$$

$$\mathrm{vOpEx}_{c,p} = \sum_f D_{f,p} \cdot P_{f,c,p} \quad (3)$$

$$\mathrm{OpEx}_{c,p} = \mathrm{fOpEx}_{c,p} + \mathrm{vOpEx}_{c,p} \quad (4)$$

where $N_{\mathrm{emp},p}$ and W are the number of full-time employees and annual working hours per employee, respectively, at a plant, estimated for each process p using H2A. $L_{c,p}$ is the hourly labor rate in country c, obtained from the Conference Board International Labor Comparisons (The Conference Board 2025), and Y_p is the annual plant output, which may be obtained from H2A hydrogen production models (NREL 2025), Aspen Plus (aspentech 2024), or Zang et al. (2024), depending on the process. The general administration, operations and maintenance, and tax rates are fixed to r_{GA}=20%, r_{OM}=4%, and r_{Tax}=2%, respectively.

The total cost per process is the sum of capital and operating expenditures:

$$C_{c,p} = \mathrm{CapEx}_{c,p} + \mathrm{OpEx}_{c,p} \quad (5)$$

Emissions modeling quantifies WTG CO_2e emissions over a 100-year timescale. The total emissions $\varepsilon_{c,p}$ per mass of fuel from a process p in country c includes upstream emissions of each input feedstock, onsite emissions $\varepsilon_p^{\mathrm{onsite}}$, and credits Credit_p for captured carbon dioxide where applicable:

$$\varepsilon_{c,p} = \sum_f \left(D_{f,p} \cdot \varepsilon_{f,c,p}\right) + \varepsilon_p^{\mathrm{onsite}} - \mathrm{Credit}_p \quad (6)$$

where $D_{f,p}$ is the demand for feedstock f per mass of fuel. The CO_2 emissions credit is obtained using mass balance:

$$\mathrm{Credit}_p = \frac{M_{\mathrm{CO2}}}{M_{\mathrm{fuel}}} \cdot C_{\mathrm{fuel}} \quad (7)$$

where M_{CO2} and M_{fuel} are the molar masses of CO_2 and the fuel, respectively, and C_{fuel} is the number of carbon atoms in a molecule of the given fuel.

Hydrogen production is modelled via five pathways: low-temperature electrolysis (LTE), steam methane reforming (SMR) without carbon capture, SMR or autothermal reforming (ATR) with carbon capture and storage (CCS) – SMR+CCS or

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ATR+CCS – and biomass gasification (BG) using LCB. Feedstock demands are obtained from Aspen Plus for the LTE pathway, H2A simulation for the BG pathway, and Zang et al. for the methane reforming pathways. Onsite emission rates are obtained from H2A and Zang et al. for the BG and methane reforming pathways, respectively, and assumed 0 for the LTE pathway. Plant CapEx is obtained from H2A for the LTE and BG pathways, and from Zang et al. for the methane reforming pathways.

Hydrogen-derived fuels considered in this study encompass ammonia, methanol, and FT diesel synthesized from hydrogen. Ammonia is produced via the Haber–Bosch process (Haber 1909), which combines hydrogen and nitrogen at high pressure and temperature. Feedstock demands, emissions, and costs for the air separation process used to obtain nitrogen are included in the Aspen Plus model used to simulate ammonia synthesis. Methanol and FT diesel are synthesized from hydrogen and CO_2 feedstocks using catalytic conversion processes, also modeled in Aspen Plus. Added costs for these conversion processes account for the plant CapEx along with labor, water, and electricity costs.

Methanol and FT diesel synthesis additionally require CO_2 as an input feedstock. The following CO_2 sources are considered: DAC, BG, SMR, and bioenergy production with carbon capture. A global DAC price of \$200/tonne CO_2 is assumed based on the range of \$125–325/tonne estimated by the IEA for a plant built today (IEA 2022). Upstream feedstock requirements and country-level emissions for DAC are evaluated based on the electricity and NG consumed by the CO_2 capture and compression processes modeled in GREET for the low-temperature adsorption process presented by Deutz and Bardow for a Climeworks facility (Deutz 2021), along with embedded emissions, NG, and water for materials reported in GREET 2024 needed to construct a Climeworks DAC facility with 100 kilotonne CO_2 capture capacity. Upstream costs and emissions of CO_2 from BG and SMR are accounted for in the modelled biogas conversion and SMR processes, respectively. A global CO_2 price from BEC of \$20/tonne is assumed based on an estimated range of \$15–30 from the IEA (IEA 2020). A global upstream emissions rate for BEC is obtained by averaging summed feedstock and fuel emissions over U.S. states using data from GREET 2024.

The total WTG costs, emissions, and feedstock demands for these fuels are calculated by combining the contributions of upstream feedstock conversion processes and the final fuel synthesis process. For a given fuel production pathway P, p_s is defined as the final fuel synthesis process, and $p \in P_{\text{upstream}}$ as the upstream feedstock conversion processes – either hydrogen production or CO_2 capture – required to supply inputs to p_s. The quantity of hydrogen or captured CO_2 from each upstream process p, required per kilogram of final fuel output based on stoichiometric requirements, is denoted $\phi_{p\to P}$. The total cost $C_{c,P}$, emissions $\varepsilon_{c,P}$, and demand $D_{f,p}$ of feedstock f per mass of fuel are then calculated as:

$$C_{c,P} = C_{c,p_s} + \sum_{p\in P_{\text{upstream}}} \phi_{p\to P} \cdot C_{c,p} \tag{8}$$

$$\text{WTG}_{c,P} = \varepsilon_{c,p_s} + \sum_{p\in P_{\text{upstream}}} \phi_{p\to P} \cdot \varepsilon_{c,p} \tag{9}$$

$$D_{f,P} = D_{f,p_s} + \sum_{p\in P_{\text{upstream}}} \phi_{p\to P} \cdot D_{f,p} \tag{10}$$

For ammonia production, P_{upstream} consists of the selected hydrogen production pathway, while p_s corresponds to ammonia synthesis via the Haber-Bosch process. For methanol and FT diesel production, P_{upstream} includes both the selected hydrogen production pathway and the selected CO_2 source, with p_s corresponding to the methanol synthesis or Fischer-Tropsch process.

LNG is assumed to be derived from fossil sources. It is produced by cooling NG feedstock to cryogenic temperatures at atmospheric pressure. The CapEx for a NG liquefaction plant is obtained from the Outlook for Competitive LNG supply (Steuer 2019) based on a large sample of projects in the U.S. lower 48 states. The emissions and feedstock demands for the liquefaction process are obtained from GREET 2024.

Conventional low-sulfur fuel oil (LSFO) is also modeled to provide a business-as-usual benchmark. The price of LSFO is estimated at \$624.50/tonne based on a one-year average of the Global 20 Ports Average over 2024 (Ship & Bunker 2025) and upstream emissions for feedstock extraction and conversion are obtained from GREET 2024.

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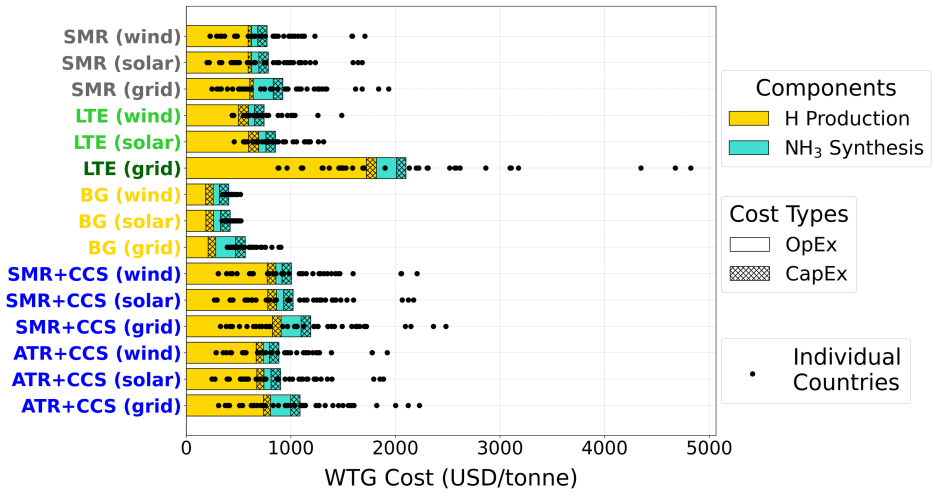


Fig. 3: WTG production costs for ammonia production pathways, broken down by CapEx and OpEx, and by production and conversion stages. Bars are averaged over all countries modeled, and total costs for individual countries are shown with black markers. Definitions of the color-coded production pathway categories listed on the y-axis can be found in Table A.1 in Appendix A.

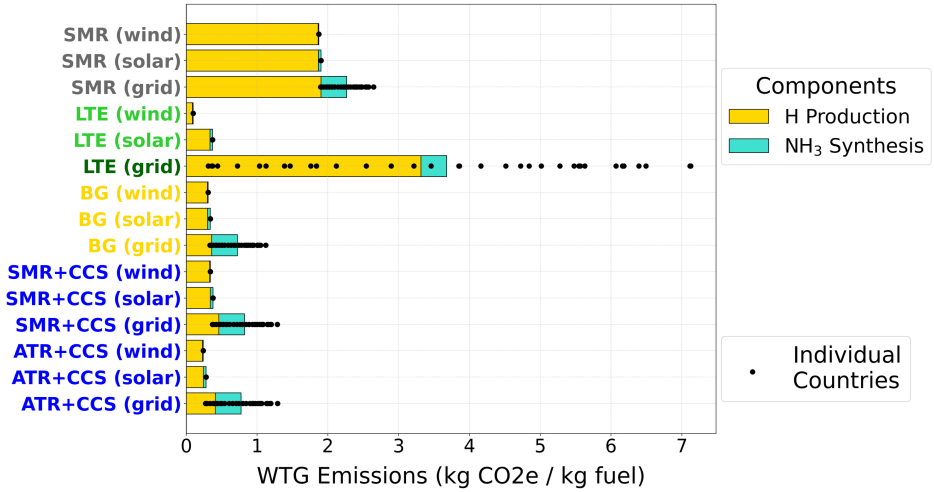


Fig. 4: WTG production emissions for ammonia production pathways. See Fig. 3 caption for details.

Global Fleet Model

The NavigaTE model (Hintze 2023) developed by the Mærsk Mc-Kinney Møller Center for Zero Carbon Shipping (MMMCZCS) is leveraged to simulate energy demand, emissions, and costs associated with operating the global fleet of container ships, bulk carriers, tankers, and gas carriers – segments that together account for over 80% of maritime greenhouse gas emissions (IMO 2021, OECD Statistics 2024). As illustrated in Fig. 5, NavigaTE integrates fuel production impacts from the fuel supply chain model to evaluate WTW results at the fleet level, considering both upstream WTG emissions from fuel production and onboard TTW CO_2 emissions from combustion of the fuel in an internal combustion engine (ICE) and, in the case of LNG vessels, methane slip in the propulsion and electrical power systems.

Fleet structure and operations are characterized in the model for individual vessel classes and sizes. Each vessel class and size, such as the Panamax size bulk carrier, is represented as a distinct fleet. Each fleet is characterized by the number of vessels, operational profiles, and power system configurations, as listed in Table B.1. Vessel specifications include technical and operational attributes such as deadweight tonnage, main engine power, fuel tank size, route characteristics, fuel properties, and powertrain technologies. Vessel-level inputs are summarized in Fig. 6.

Vessel operations are represented using annual distributions. The route for each fleet is characterized by the average number of port calls, the fraction of time spent at sea versus in port, and distributions of vessel speeds and cargo utilization rates.

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The annual energy demand of each fleet is calculated in the NavigaTE model based on vessel-specific operational parameters. For vessels in each fleet, the NavigaTE model calculates the annual useful energy demand based on operational parameters. The total annual energy demand, denoted E_{annual}, is the sum of energy consumed during sea operations and port operations:

$$E_{\text{annual}} = E_{\text{sea}} + E_{\text{port}} \tag{11}$$

The energy demand at sea, E_{sea}, is calculated on an annual basis (t_{year}) as:

$$E_{\text{sea}} = \left[\sum_k P_{\text{prop}}(s_k, u_k) f_k + P_{\text{elec},s} + P_{\text{heat},s} \right] \cdot \left(f_s \cdot t_{\text{year}} \right) \tag{12}$$

where f_s is the fraction of time the vessel spends at sea. $P_{\text{prop}}(s_k, u_k)$ is the operating power of the main propulsion system under sailing condition k, which is characterized by the vessel speed s_k and fractional cargo space utilization u_k. f_k represents the portion of time at sea that the vessel spends in condition k. $P_{\text{elec},s}$ and $P_{\text{heat},s}$ represent the power demands of the electrical and heating systems while the vessel is at sea.

The energy demand at port is calculated similarly, with the propulsion power component excluded:

$$E_{\text{port}} = \left[P_{\text{elec},p} + P_{\text{heat},p} \right] \cdot \left(f_p \cdot t_{\text{year}} \right) \tag{13}$$

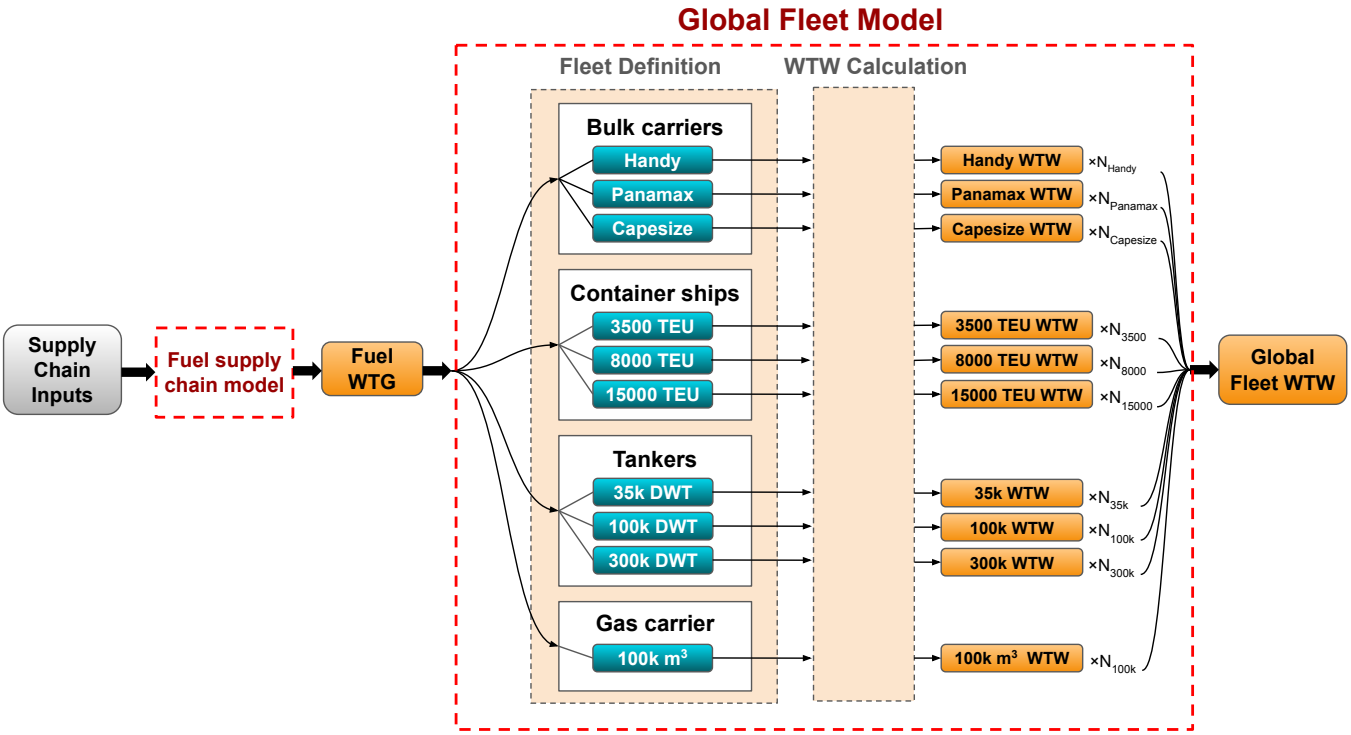


Fig. 5: Illustration of how the NavigaTE global fleet model interfaces with the WTG results produced by the fuel supply chain model to evaluate WTW emissions for the global fleet of bulk carriers, container ships, tankers, and gas carriers

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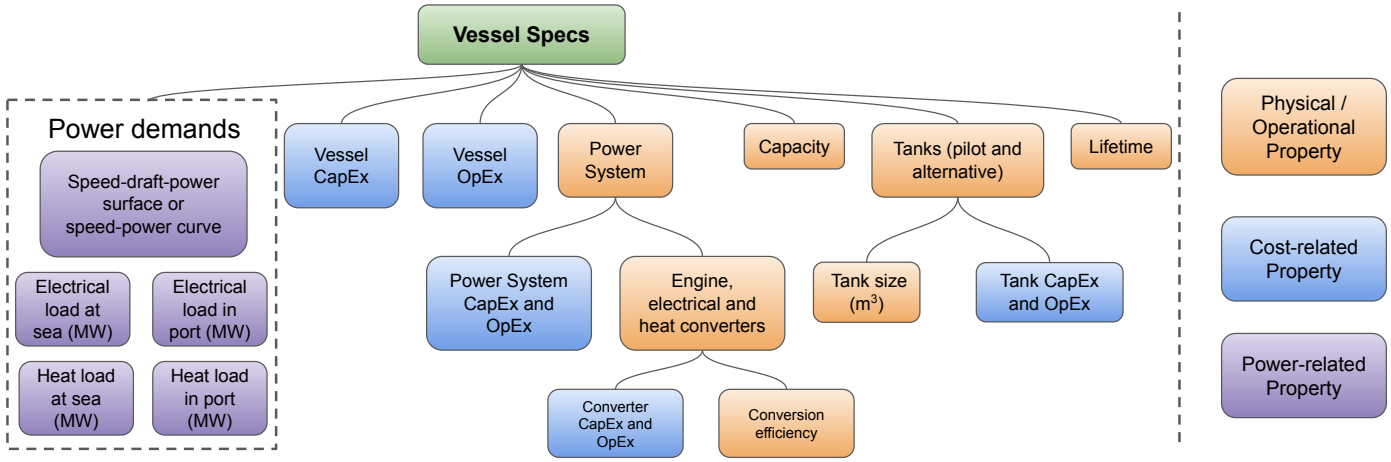


Fig. 6: Summary of specifications input to the NavigaTE model to define a given vessel class.

The fuel demand and onboard thermal conversion efficiency for each vessel determine its annual fuel mass requirements, $M_{\text{annual},F}$, for each fuel $F \in$ {main fuel, pilot fuel}. This accounts for the fuel's lower heating value LHV_F, , along with onboard thermal conversion efficiencies η_x, where x indexes the major energy uses: propulsion, electricity, and heating. This requirement is calculated as:

$$M_{\text{annual},F} = \sum_x \phi_F \frac{E_x}{\text{LHV}_F \cdot \eta_x} \quad (14)$$

where ϕ_F is the energy share of fuel F in the vessel's tanks. For ammonia, methanol, and LNG vessels equipped with dual-fuel systems, which can operate on a main fuel alongside pilot LSFO fuel, the pilot fuel energy share is fixed to values of 5% for ammonia and methanol and 1% for LNG to ensure that the majority of energy is supplied by the alternative fuel.

Feedstock consumption for fuel production, including electricity, water, biomass, and captured carbon dioxide, is incorporated from the fuel supply chain model. The annual consumption of each feedstock f is computed as:

$$\text{Feedstock}_{\text{annual},f,P} = \sum_{F \in \{\text{main,pilot}\}} \left(M_{\text{annual},F} \cdot D_{f,P}\right) \quad (15)$$

where $D_{f,P}$ is the feedstock demand per mass of fuel produced via pathway P (see Eq. 10).

Costs and emissions at the vessel level are computed on an annual basis, accounting for both WTW CO_2e emissions and total operational costs. For a given fuel F and production pathway P in country c, WTW emissions include both the WTG upstream emissions $\text{WTG}_{c,P,F}$ and tank-to-wake (TTW_F) onboard combustion emissions per mass of fuel:

$$\text{WTW}_{c,P,F} = \sum_{F \in \{\text{main,pilot}\}} (\text{WTG}_{c,P,F} + \text{TTW}_F) \cdot M_{\text{annual},F} \quad (16)$$

The total annual vessel cost includes capital expenditures (CapEx), operational expenditures (OpEx), and fuel costs:

$$C_{\text{annual}} = \text{CapEx}_{\text{annualized}} + \text{OpEx}_{\text{annual}} + \sum_F C_{c,P,F} \cdot M_{\text{annual},F} \quad (17)$$

CapEx is annualized over the vessel lifetime T, accounting for financing costs:

$$\text{CapEx}_{\text{annualized}} = \frac{\text{CapEx}}{T} + \text{CapEx} \cdot C_{\text{capital}} \quad (18)$$

where C_{capital} is the cost of capital, assumed to be 5.5%.

Fleet-level Aggregation

The fleet-level aggregation builds on vessel-level results by summing outputs across all vessels within each fleet, defined by a given vessel class and size (e.g., Panamax bulk carrier). Vessel-level results are aggregated to each fleet as follows:

$$\text{Fleet Result}_{\text{class,size}} = N_{\text{class,size}} \cdot \text{Result}_{\text{class,size}} \quad (19)$$

Here, $N_{\text{class,size}}$ is the number of vessels of a given class and size in the global fleet, and $\text{Result}_{\text{class,size}}$ is the per-vessel result of interest – either emissions, feedstock consumption, or costs. This can be further aggregated to obtain results for the full global fleet:

$$\text{Global Result} = \sum_{\text{class}} \sum_{\text{size}} \text{Fleet Result}_{\text{class,size}} \quad (20)$$

Cost and Emissions Metrics

Comparing alternative marine fuels requires joint consideration of both cost and emissions performance. Traditionally, the Cost of Abated Carbon (CAC) has served as the standard metric for such comparisons (Weyant 1993), expressing the incremental cost of a fuel relative to low-sulfur fuel oil (LSFO) per unit of

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carbon dioxide equivalent (CO_2e) emissions avoided:

$$\mathrm{CAC} = \frac{\mathrm{Cost}_{\text{alt fuel}} - \mathrm{Cost}_{\mathrm{LSFO}}}{\mathrm{Emissions}_{\mathrm{LSFO}} - \mathrm{Emissions}_{\text{alt fuel}}} \tag{21}$$

While intuitive, CAC presents significant drawbacks in the context of marine fuels. For fuel pathways that increase emissions relative to LSFO, the metric becomes ambiguous and difficult to interpret. Furthermore, CAC values become unstable when emissions reductions are small, leading to exaggerated results that obscure meaningful comparisons across pathways and regions.

Given these limitations, this study introduces the "Average Cost and Emissions Ratio" R_{av} as an alternative evaluation metric. This formulation compares each alternative fuel to LSFO by averaging the ratio of annual cost and emissions to operate the global fleet on the alternative fuel versus LSFO:

$$R_{\mathrm{av}} = \frac{1}{2}\left(\frac{\mathrm{Cost}_{\text{alt fuel}}}{\mathrm{Cost}_{\mathrm{LSFO}}} + \frac{\mathrm{Emissions}_{\text{alt fuel}}}{\mathrm{Emissions}_{\mathrm{LSFO}}} \right) \tag{22}$$

The use of R_{av} offers several advantages for this work. First, it is always well-defined, regardless of whether an alternative fuel increases or decreases emissions relative to LSFO. Second, it enables clear benchmarking, with LSFO having a reference value of 1, providing an intuitive basis for comparison across pathways. Third, it transparently reflects both economic and environmental performance without distortion from extreme or ambiguous values. Finally, the formulation allows flexibility for future applications, including weighted versions that can emphasize either cost or emissions in line with specific policy or industry objectives.

RESULTS AND DISCUSSION

Feedstock Requirements

Feedstock requirements to scale up fuel production to meet the current demand of the global maritime fleet are highly pathway-dependent. Figs. 7 and 8 show requirements across all fuels and pathway classes for two sample feedstocks, electricity and NG.

Across the fuels and pathways considered, electrolytic (LTE) routes consume 8-60x more electricity than other pathways for methanol, 4-60x more for FT diesel, and 7-9x more for ammonia. Scaling to the global fleet, this translates to a total electricity requirement ranging from 124 terawatt-hours (TWh) to 6,858 TWh per year depending on the production pathway, representing up to 23% of total global electricity consumption in 2022 (EIA 2024a). This underscores the urgency of expanding low carbon power generation to enable largescale electrolytic fuel production for maritime applications. This expansion must also meet the concurrent rise in electricity demand from other electrifying sectors (IEA 2024).

Natural gas consumption, shown in Fig. 8, also emerges as an important consideration. For pathways that rely on methane reforming (SMR or ATR, with or without CCS), NG demand could reach up to 34 exajoules (EJ) annually for FT diesel production, equal to approximately 24% of global NG consumption in 2022 (IEA 2024b). This scale of demand would place significant pressure on existing NG supplies. Obtaining CO_2 from DAC for methanol and FT diesel production is also seen to consume significant NG – 8.7 EJ for FT diesel and 3.9 EJ for methanol, due to the assumption in GREET that thermal heat for the process is supplied by NG combustion.

CO_2 supply also becomes an important constraint for carbon-containing fuels. Methanol and FT diesel synthesis requires large volumes of CO_2, obtained either from LCB or direct air capture (DAC). Across modelled pathways, global CO_2 demand reaches 576 million tonnes for methanol and 1,287 million tonnes for FT diesel. These levels represent 2-5x the 272 million tonnes of CO_2 projected to be captured and used in 2025 (IEA 2019) – primarily for fertilizer and urea production and enhanced oil recovery – pointing to a massive scale-up need in carbon capture and CO_2 handling infrastructure, especially given growing competition for CO_2 for other end-uses such as carbon sequestration. Brynolf et al. (2022) and Korberg et al. (2021) similarly caution that limited CO_2 and biomass supplies could become bottlenecks, reinforcing a need for regional planning and DAC supply certainty.

Biomass availability constraints also become relevant for carbon-containing fuels. For pathways that obtain hydrogen or CO_2 from BG, modeled LCB demand can reach 1.6 Gt for FT diesel – equivalent to 25–31 EJ using typical LCB lower heating values (Matin 2023). Comparing with sustainable availability of biomass from forestry residues, estimated at around 3 to 26.1 EJ annually (Daehn 2025), relying on residues alone would likely be insufficient to meet this demand. Meeting these requirements would likely necessitate dedicated energy crops, raising further considerations around sustainable land use. This adds quantitative support to concerns raised by Foretich et al. (2021) and Xing et al. (2021) around constraints and logistical challenges of acquiring biogenic feedstocks for these fuels.

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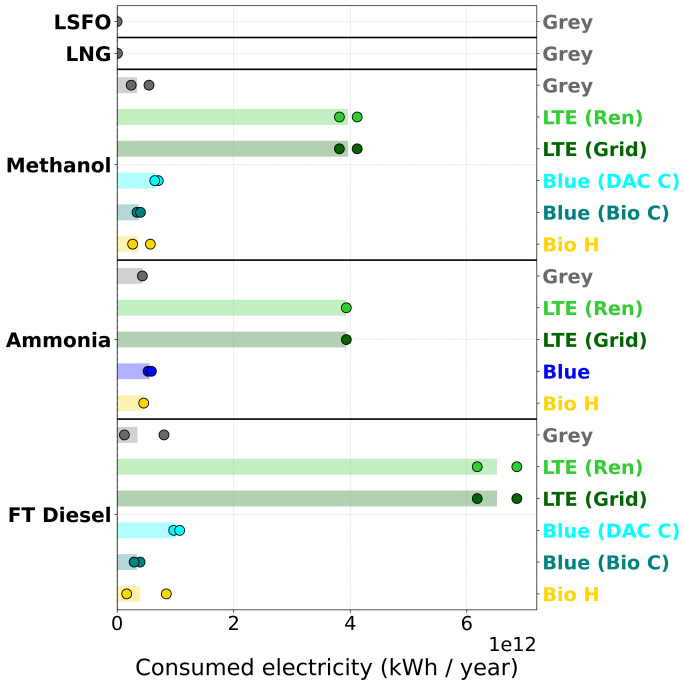


Fig. 7: Modelled annual electricity consumption to supply the global fleet for all classes of production pathways considered. Markers represent individual pathways, and bars extend to the average value over all pathways belonging to the given category (see Table A.1 for pathway category definitions).

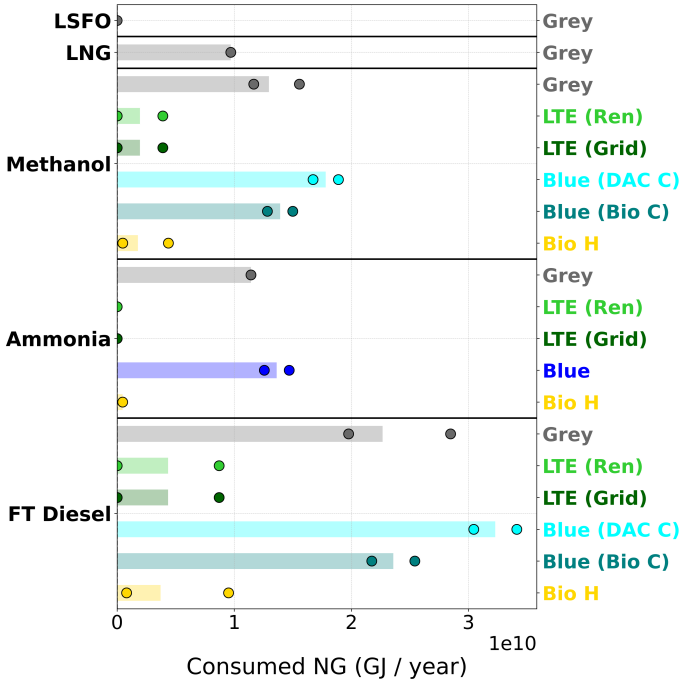


Fig. 8: Modelled annual NG consumption to supply the global fleet for all classes of production pathways considered. See Fig. 7 caption for details.

Cost and Emissions Trade-offs

A comparison of costs and emissions involved in operating the global fleet on alternative fuels reveals a similarly complex landscape, with significant variation across fuels, production pathways, and geographies. The violin plots in Figs. 9 and 10 compare the ranges of total annualized cost and WTW emissions to operate the global fleet between each fuel, and each production pathway category defined in Table A.1.

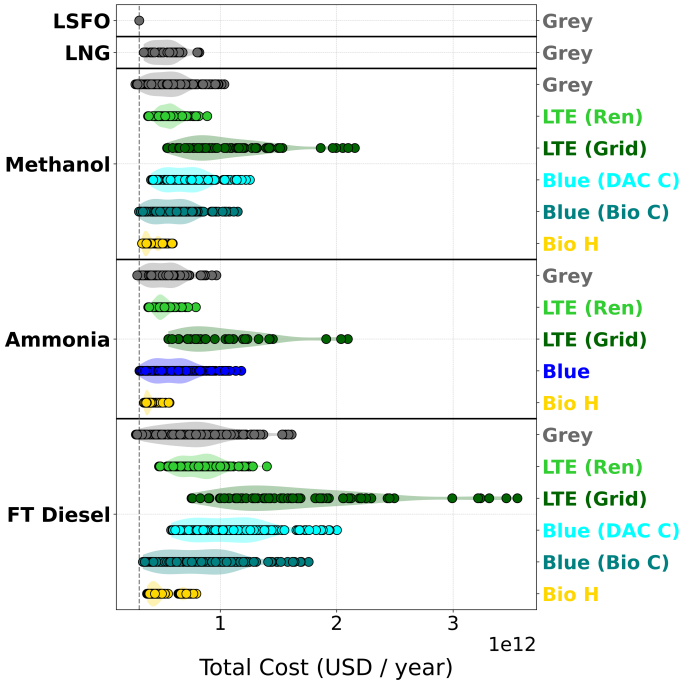


Fig. 9: Comparison of total annual cost to operate the global fleet between modelled fuels, production pathway categories, and regions. Individual points represent specific pathways and countries, and violin widths are proportional to point density.

Cost and emissions breakdowns by stage also help to identify and compare the important drivers of observed differences between fuels and pathways. For example, Fig. 11 shows country-averaged cost breakdowns for grid-powered FT diesel pathways, and Fig. 12 the corresponding WTW emissions (wind- and solar-powered pathways are excluded for brevity). Fuel production dominates the total annual global fleet cost for all pathways. For pathways using DAC or BEC carbon, WTG emissions are often negative. FT diesel and methanol achieve only modest TTW CO_2 reductions (19% and 11%, respectively), so overall WTW benefits hinge on low upstream emissions.

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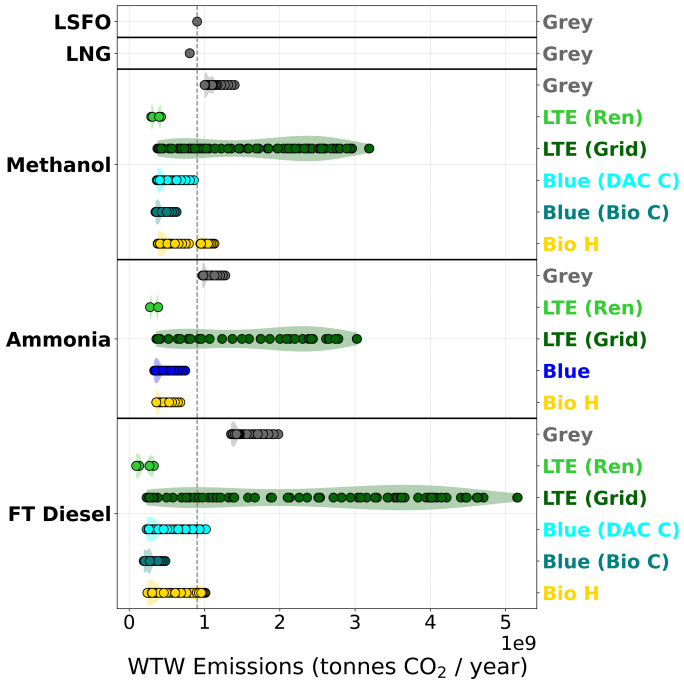


Fig. 10: Comparison of total annual WTW emissions to operate the global fleet between modelled fuels, production pathway categories, and regions. See Fig. 9 caption for details.

For ammonia-powered dual-fuel vessels, modeled TTW emissions are 95% lower than LSFO vessels – with the remaining 5% from LSFO combustion as pilot fuel – enabling more WTG-positive pathways to achieve net WTW improvements over LSFO. As Fig. 10 shows, ammonia emissions exceed LSFO only for methane reforming pathways without CCS and for LTE pathways in countries with high-carbon grids such as India, South Africa, and Indonesia.

Several important pathway-level trends are observed from these comparisons. Across fuels, blue and electrolytic hydrogen pathways with renewable power sources both have potential to achieve up to a 90% reduction in annual WTW emissions for the global fleet relative to LSFO, and hydrogen from BG can achieve up to 80% reduction. When paired with biogenic or DAC-sourced carbon, our modeling finds that ammonia, methanol, and FT diesel produced via these hydrogen production pathways can support deep decarbonization of maritime shipping.

Hydrogen production via steam methane reforming (SMR) without carbon capture performs poorly across all fuels and countries, resulting in WTW emissions that exceed those of conventional low-sulfur fuel oil (LSFO). These pathways also tend to increase the total cost of operating the global fleet, offering neither emissions reductions nor cost benefits. While SMR without carbon capture could serve as a short-term transition option as lower-carbon alternatives are built out, the addition of carbon capture significantly improves performance. With carbon sequestration, WTW emissions fall below LSFO levels for nearly all fuels and countries. Importantly, the incremental cost of adding carbon capture is relatively modest compared to the regional variation in methane reforming costs, suggesting that strategic siting of production in low-cost regions can easily offset the expense of including carbon capture.

Using NG directly as a fuel in the form of fossil LNG provides only marginal emissions benefits relative to conventional low-sulfur fuel oil (LSFO). This pathway also increases costs in most production countries, making it a poor candidate as a transitional fuel. Xing et al. (2021) likewise caution against large-scale LNG adoption, citing limited carbon-savings and infrastructure lock-in. Some potential may exist for LNG to serve as a bridge to low-carbon methane alternatives, such as power-to-methane (Blanco 2018) or methane produced via anaerobic digestion (Ali 2024), though upstream methane leakage will be important to monitor and control regardless of the production mechanism.

Methane reforming pathways that use hydrogen from SMR+CCS or ATR+CCS – 'Blue (DAC C)' and 'Blue (Bio-C)' – offer meaningful emissions reductions relative to LSFO, provided the captured carbon is permanently sequestered. Economically, methane reforming pathways with CCS tend to be more expensive than LSFO, except in a few countries endowed with low-cost NG and electricity.

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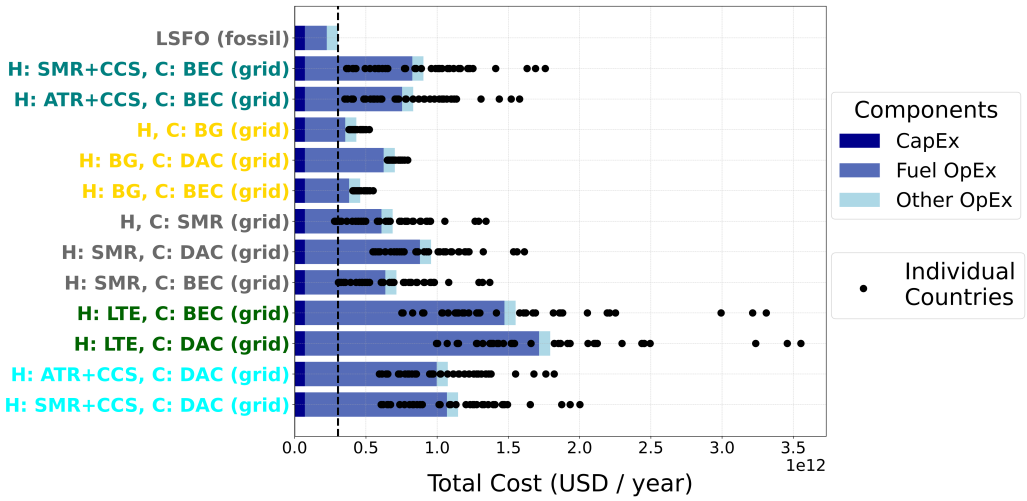


Fig. 11: Total annualized cost to operate the global fleet on FT diesel for all modeled production pathways powered by grid electricity. The total cost for each country is shown as a black dot, and the average over all countries is shown as a stacked bar.

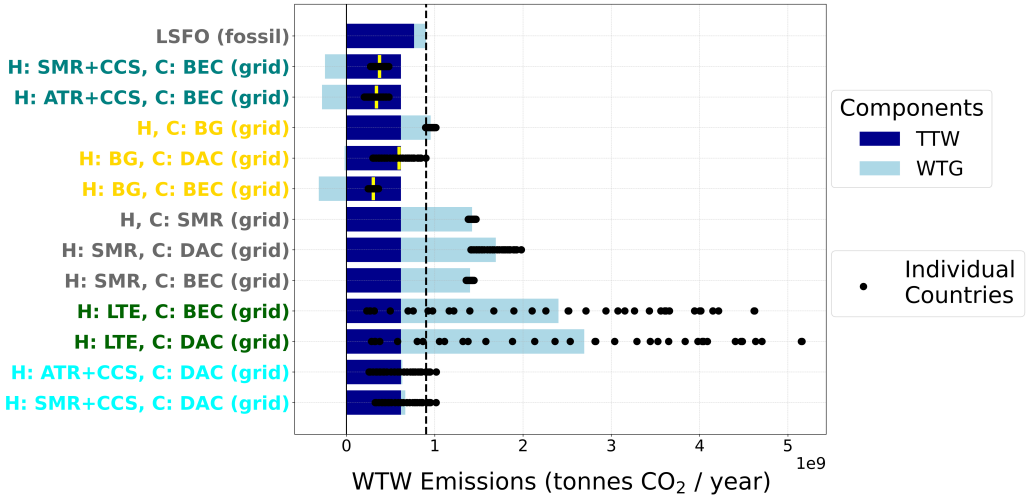


Fig. 12: Total WTW emissions to operate the global fleet on FT diesel for all modeled production pathways powered by grid electricity. The WTW emissions for each country is shown as a black dot, and the average over all countries is shown as a stacked bar. For pathways with negative WTG emissions, a yellow band is drawn at the stacked total.

Regional cost and emissions differences between fuel production regions are seen to be just as important as fuel- and pathway-level choices. Depending on the pathway, costs and emissions to operate the global fleet on alternative fuels can vary by up to a factor of ten depending on the production country. This variation is especially pronounced for electrolytic hydrogen pathways that rely on grid electricity, where regional differences in grid carbon intensity and electricity price dominate the analysis.

The average cost and emissions ratio R_{av}, shown as a violin plot in Fig. 13, helps to compare trade-offs between added costs and emissions benefits between production regions. Shown geospatially in Figs. 14 and 15 for grid electrolytic and blue ammonia pathways, respectively, this metric can vary by over 5x between different regions for the same production pathway, with relatively good performance for grid electrolytic pathways in regions like Brazil, Scandinavia, and Canada with relatively clean

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and low-cost grid power, and for methane reforming pathways in regions like Russia, Saudi Arabia, and Canada with low-cost NG.

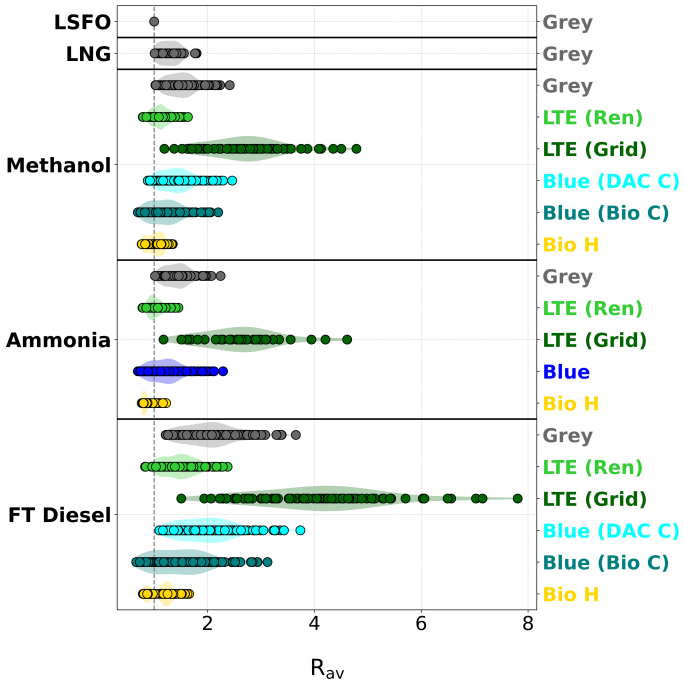


Fig. 13: Comparison of the average cost and emissions ratio R_{av} evaluated for the full global fleet between modelled fuels, production pathway categories, and regions. See Fig. 9 caption for details.

Across all fuels, pathways that use hydrogen produced from LCB (Bio H) and LTE powered by wind or solar can yield $R_{av}<1$ in many countries, indicative of improved cost-emissions performance relative to conventional LSFO. For solar-powered LTE pathways, $R_{av}<1$ is seen in most southeast Asian countries, particularly China, as well as Australia, Saudi Arabia, and parts of southern Europe that benefit from low-cost solar. For wind-powered LTE, the lowest ratios are seen in China and Brazil followed by Spain and the UK, and are highest in Japan and island nations in Southeast Asia where wind generation is typically more expensive. No grid-powered LTE pathways yet have low enough costs and emissions to quite achieve $R_{av}<1$, but competitive values are seen in countries like Norway and Sweden that currently generate over 50% of their energy from renewable sources (Energy Institute 2024), as shown in Fig. 14 for ammonia. For methanol and FT diesel, LTE pathways with $R_{av}<1$ are only seen in cases where the CO_2 used to produce the fuel comes from BEC, as obtaining CO_2 from methane reforming increases WTW emissions relative to LSFO, and the higher assumed cost of obtaining CO_2 from DAC typically increases fleet operations by approximately 25% relative to BEC.

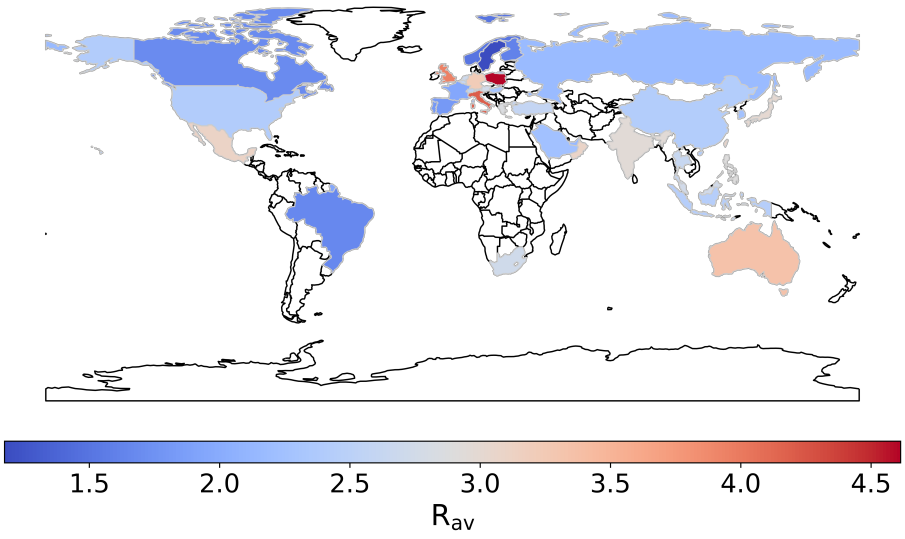


Fig. 14: Comparison of the average cost and emissions ratio R_{av} to operate the global fleet on ammonia between modelled countries for hydrogen produced via low-temperature electrolysis using grid power.

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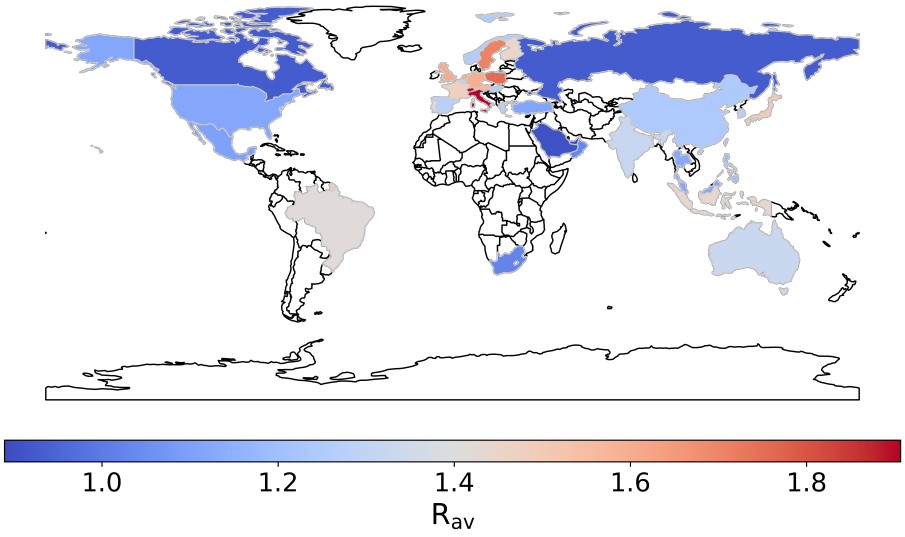


Fig. 15: Comparison of the average cost and emissions ratio R_{av} to operate the global fleet on ammonia between modelled countries for "blue" ammonia produced from auto-thermal reforming with carbon capture and storage using grid power.

Among pathways that generate hydrogen from methane reforming, those that capture and sequester the carbon from the methane reforming process – "Blue" pathways for ammonia or "Blue (Bio/DAC C)" for methanol and FT diesel – can achieve R_{av} <1 for countries such as Russia, Saudi Arabia, and Canada with low-cost NG, as seen in Fig. 15.

The R_{av} metric also helps identify fuels and pathways that best balance cost and emissions trade-offs across production regions. From Fig. 13, pathways using renewable electrolytic and biogenic hydrogen are seen as the best-performing options across fuel types and regions. While these pathways generally incur higher costs than conventional fuels, they deliver the greatest emissions reductions. However, each come with important feedstock constraints that will need to be accounted for – global wind and solar electricity generation would need to increase by 85% (ammonia) to 150% (FT diesel) relative to 2024 (Ember 2025) to supply sufficient fuel for the global fleet via renewable LTE pathways and, as discussed earlier, acquisition of sustainable LCB feedstocks will become an increasingly important challenge as demand for alternative fuels grows. Meanwhile, methane reforming with CCS can provide competitive performance in select geographies, but generally at a higher cost premium and with significant demands on the global NG supply.

Across production pathways, ammonia is generally found to provide the greatest emissions reductions for the global fleet at least cost, as indicated by its relatively low R_{av} values in Fig. 13, though methanol has only slightly worse performance for most pathways. These findings corroborate and generalize the corridor-specific results of Douglas et al. (2025) to a global context; in their study of bulk iron ore carriers, ammonia outperformed methanol in terms of the cost of abated carbon. However, we find methanol to yield slightly higher WTW emissions than ammonia, in contrast to Douglas et al. This divergence likely reflects differences in assumptions around non-CO_2 emissions: while Douglas et al. model NO_x and N_2O formation and aftertreatment in detail, our analysis only accounts for CO_2 from pilot fuel combustion in TTW emissions. This is a significant simplification due to the large global warming potential of N_2O and highlights the need emphasized by Chiong et al. (2021) for further empirical research to extend detailed specific assessments of non-CO_2 impacts of alternative fuels like that by Douglas et al. to more general contexts. Recent work by Wong et al. (2024) provides a valuable foundation, modeling global air quality and health impacts of NO_x, ammonia, and N_2O emissions from a global ammonia-powered fleet across engine and scrubbing configurations.

While further work is also needed to fully capture added capital and operational costs of storing and combusting liquefied

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ammonia, the relatively small contribution of these components compared with fuel costs suggests this refinement is unlikely to fundamentally alter the conclusions of this study. While our modeling does not assess safety, handling, or infrastructure requirements, these factors highlighted by Foretich et al. (2021) remain critical for the practical deployment of ammonia and merit further integration into future comparative assessments. Among carbon-containing fuels, methanol generally performs better than FT diesel in terms of cost, WTW emissions, and feedstock efficiency. Despite its less favorable profile, FT diesel may still play an important role as a drop-in fuel while methanol- and ammonia-ready vessels are being deployed.

CONCLUSIONS

This study assesses the viability of ammonia, methanol, and FT diesel as low-carbon maritime fuels by integrating detailed models of their production and fleet-wide adoption. The results highlight critical trade-offs in emissions, costs, and feedstock demands across fuels, pathways, and regions.

Feedstock requirements vary widely by pathway. Electrolytic fuels demand substantial electricity – up to 23% of present-day global generation – while FT diesel and methanol production require large quantities of CO_2 and lignocellulosic biomass. Projected global CO_2 demand reaches up to 1,300 Mt, representing 2–5 times the volume of CO_2 projected to be captured and utilized globally in 2025. Biomass demand for FT diesel production exceeds the estimated sustainable availability of lignocellulosic biomass from forestry residues. Blue hydrogen pathways also entail substantial natural gas use, with fleet-scale demand reaching up to 24% of global consumption. These constraints underscore the importance of aligning fuel deployment with feedstock availability and long-term scalability.

Cost and emissions outcomes are similarly diverse. Renewable electrolytic and blue hydrogen pathways can achieve up to 90% emissions reductions but are often more costly than fossil-derived fuels. Pathways that use hydrogen from BG can achieve up to 80% emissions reduction, but often at lower cost. Methane reforming without carbon capture generally performs worse than LSFO in both emissions and cost. The addition of carbon capture can reduce emissions below LSFO levels if storage is permanent, but these blue pathways remain costlier than renewable electrolytic and biogenic options in most countries.

Geographic variation plays a central role. For the same fuel and pathway, emissions and costs differ by up to a factor of ten across countries, driven by local electricity prices, grid emissions intensity, and other feedstock costs. The average cost and emissions ratio R_{av}, introduced in this study, provides a robust metric for comparing these trade-offs and identifying favorable production regions.

Among fuels, ammonia offers the most consistent emissions reductions at relatively low cost, while methanol performs nearly as well and faces fewer infrastructure hurdles. FT diesel generally ranks lower on both cost and emissions tradeoffs and feedstock efficiency but may still serve as a valuable transitional option.

These findings support a regionally tailored approach to fuel deployment that accounts for both technical performance and local resource conditions. Further work is needed to evaluate future cost and emissions trajectories, quantify and compare global non-GHG impacts in addition to GHG, and refine engineering estimates of vessel-level trade-offs such as added costs and cargo loss due to storage of low-density and/or cryogenic fuels.

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APPENDIXES

Appendix A: Summary of Pathway Classes

Table A.1: Fuel production pathway classes and their descriptions

Categories	Description
Grey	Fossil derived pathways without carbon capture: LSFO, LNG, and hydrogen-derived pathways with hydrogen from SMR.
Blue	Non-carbon-containing fuel pathways in which the hydrogen is obtained from SMR+CCS or ATR+CCS.
Blue (Bio C)	Carbon-containing fuels use hydrogen from SMR+CCS or ATR+CCS; captured carbon is sequestered, and biogenic carbon (BG or BEC) is used in fuel production.
Blue (DAC C)	Carbon-containing fuels use hydrogen from SMR+CCS or ATR+CCS; captured carbon is sequestered, and DAC-derived carbon is used in fuel production.
LTE (Grid)	Hydrogen from LTE powered by grid electricity.
LTE (Ren)	Hydrogen from LTE powered by solar or wind.
Bio H	Hydrogen from BG. For carbon-containing fuels, carbon may come from BG or other sources.

Appendix B: Fleet Model Details

Table B.1: Summary of fleets defined in the NavigaTE simulation, along with the number of vessels in each fleet.

Vessel Class	Vessel Size	Vessels
Bulk carrier	Capesize	2,067
Bulk carrier	Panamax	3,110
Bulk carrier	Handy	6,929
Container Ship	15,000 TEU	808
Container Ship	8,000 TEU	1,043
Container Ship	3,500 TEU	3,983
Tanker	300k DWT	1,529
Tanker	100k DWT	1,643
Tanker	50k DWT	9,770
Gas carrier	100k m^3	2,156

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Table B.2: Specifications used to define each fleet

Specification	Description
Vessels	Vessel and power system combinations included in the fleet. Each fleet contains one vessel class and size but may include multiple power system types.
Initial Vessels	Number of vessels in the fleet at the initial time step.
Initial Split	Distribution of vessels between power systems at the initial time step.
Initial Ages	Distribution of vessel ages by power system, typically ranging from 0 to 24 years.
Trade Growth	Projected annual growth in cargo miles carried by the fleet (used only in future projections).
Order Books	Newbuild vessel orders per power system in each year (used only in future projections).

Table B.3: Vessel power systems defined in the NavigaTE simulation

Power System	Fuel Mode	Main Fuel	Pilot Fuel
Ammonia	Dual-fuel	Ammonia	LSFO
Methanol	Dual-fuel	Methanol	LSFO
Methane	Dual-fuel	LNG	LSFO
Diesel	Single-fuel	FT Diesel	None
LSFO	Single-fuel	LSFO	None

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